DATA SCIENCE CAPSTONE

Predicting Falcon 9 first-stage landings

From launch records to an interpretable machine-learning comparison

IBM Applied Data Science Capstone



Executive summary

90

Falcon 9 launches in the 2010–2020 IBM snapshot

66.7%

Overall successful first-stage landing rate

83.3%

Best observed test accuracy on 18 held-out rows

- Methods: API + web scraping → wrangling → EDA/SQL → Folium/Dash → classification.
- Result: landing performance improved strongly over time; KSC LC-39A led sites at 77.3%.
- Model result: KNN, SVM, and logistic regression tied on test accuracy; KNN won the CV tie-break.

Why predict a first-stage landing?

Reusable boosters can change launch economics — but only when the first stage returns safely.

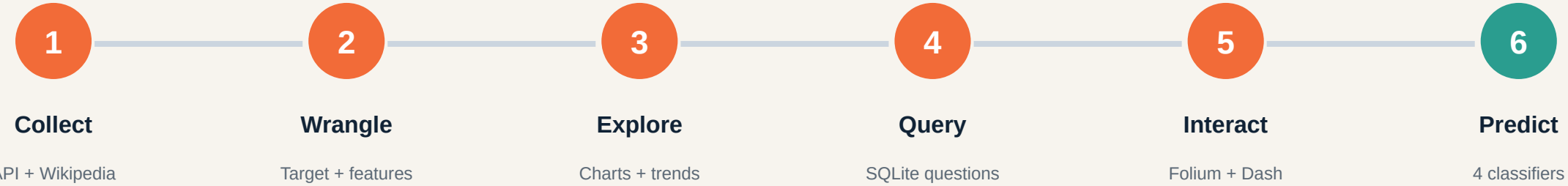
- Target: Class = 1 for a successful landing
- Scope: 90 launches from June 2010 to November 2020
- Goal: interpretable comparison, not operational certification

Research question

Can historical mission, payload, launch-site, orbit, and booster information predict a successful Falcon 9 first-stage landing?

METHODS

End-to-end methodology



The same frozen course snapshot feeds every analysis layer, allowing figures, queries, maps, dashboard views, and model results to remain consistent.

Repository https://github.com/ScottPhantom/SpaceX-Capstone-Project

Data collection — SpaceX API

1

Request

GET past launches

2

Normalize

Flatten nested JSON

3

Enrich

Rocket, pad, payload, core

4

Filter

Falcon 9, cutoff date

Key fields

flight number · date · booster · payload mass · orbit · launch site · landing outcome

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Data collection — web scraping

Wikipedia launch tables provide a flat, human-readable cross-check for mission metadata.

1. **Read HTML tables**
2. **Detect Flight No. columns**
3. **Concatenate launch rows**
4. **Export reproducible CSV**

Implementation: pandas.read_html() with an explicit table-selection guard.

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Data wrangling and feature engineering

- Standardize dates and numeric payload mass; impute missing payload mass with the sample mean.
- Map landing outcomes into Class: 1 = successful landing, 0 = unsuccessful landing.
- One-hot encode launch site, orbit, booster serial, grid fins, legs, and reuse indicators.
- Verify the 83-column model matrix contains no missing values before training.

90
rows retained

83
encoded features

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METHODOLOGY

EDA with visualization

Each chart answers a specific operational question.

Site success bar	Which launch site has the strongest observed landing rate?
Payload scatter	How do payload mass and site relate to landing outcome?
Yearly trend	Did landing performance improve over time?
Orbit summary	Which mission profiles show stronger or weaker outcomes?

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METHODOLOGY

EDA with SQL

A SQLite copy of the 101-row lab table supports transparent, repeatable questions.

1

Load

CSV → SPACEXTBL

2

Filter

LIKE, BETWEEN, dates

3

Aggregate

COUNT, SUM, MIN, MAX

4

Rank

GROUP BY + ORDER BY

Questions cover sites, NASA CRS payload, successful landings, maximum payload boosters, failure timing, and launch-site rankings.

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METHODOLOGY

Interactive visual analytics

FOLIUM

Geospatial questions

- Where are the launch sites?
- Which records succeeded or failed?
- How close are sites to infrastructure?

PLOTLY DASH

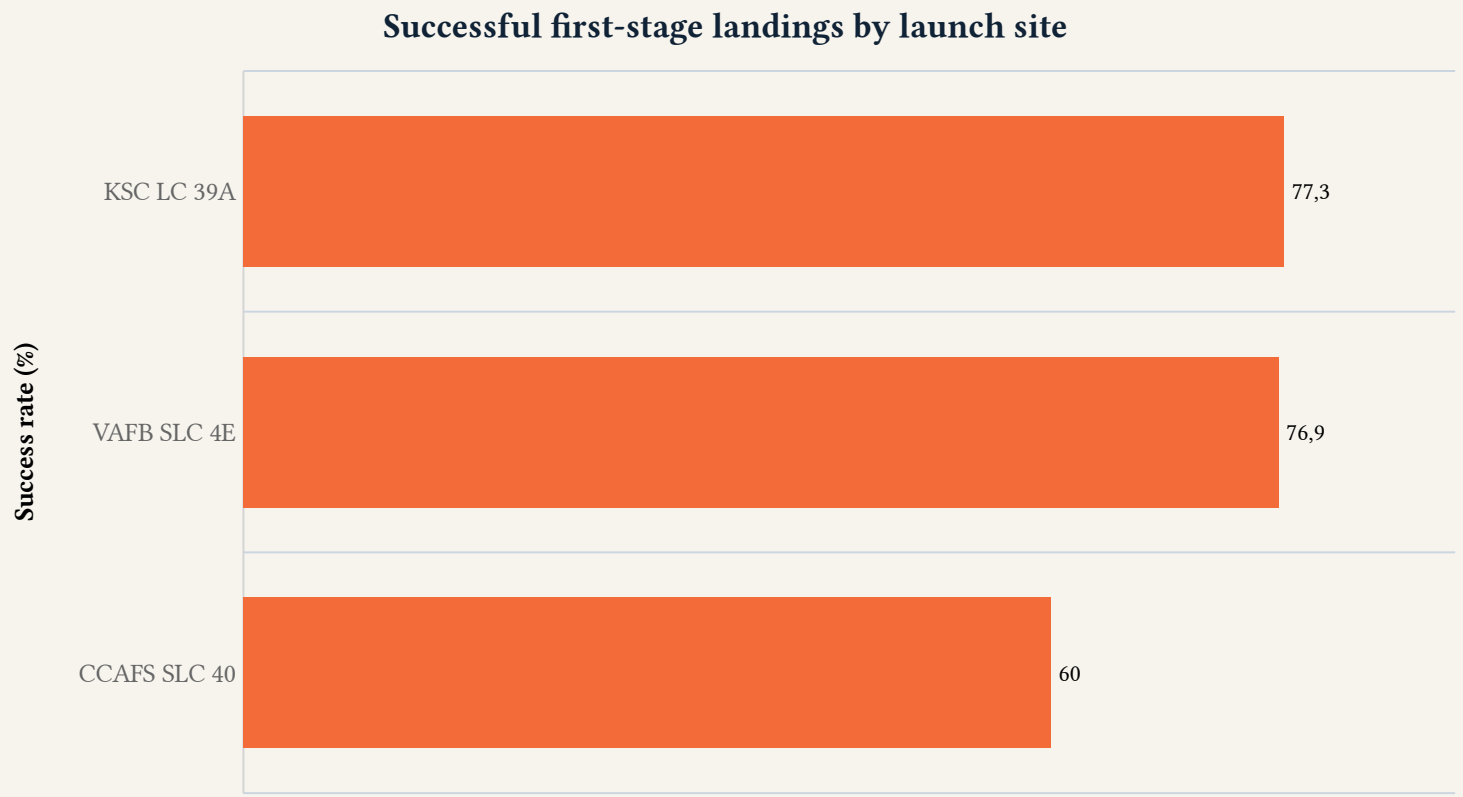
Interactive filtering

- Select one site or all sites
- Filter payload mass range
- Update pie and scatter views together

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EDA RESULTS

KSC LC-39A leads observed site success



77.3%

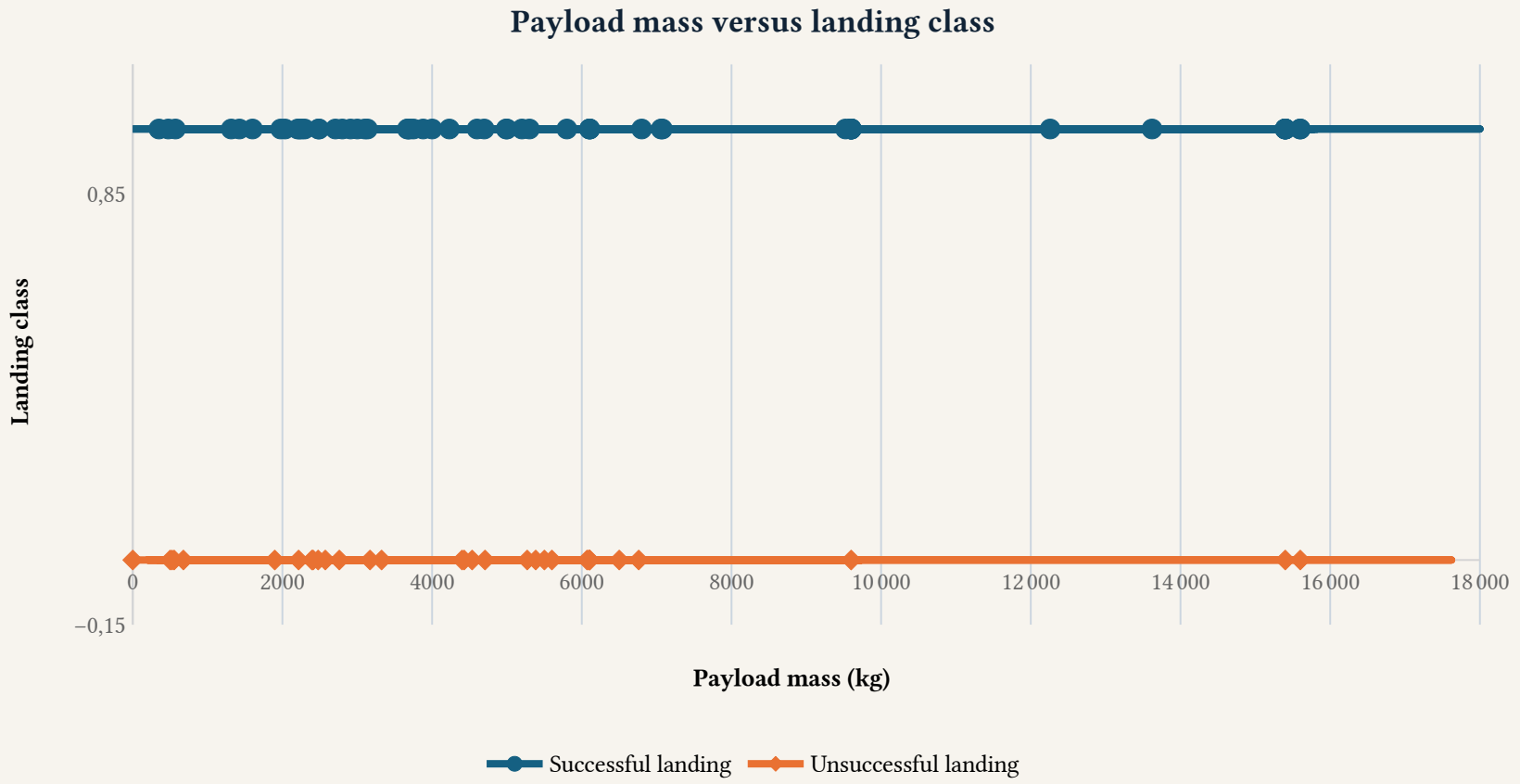
KSC LC-39A · 22 launches

60.0%

CCAFS SLC-40 · 55 launches

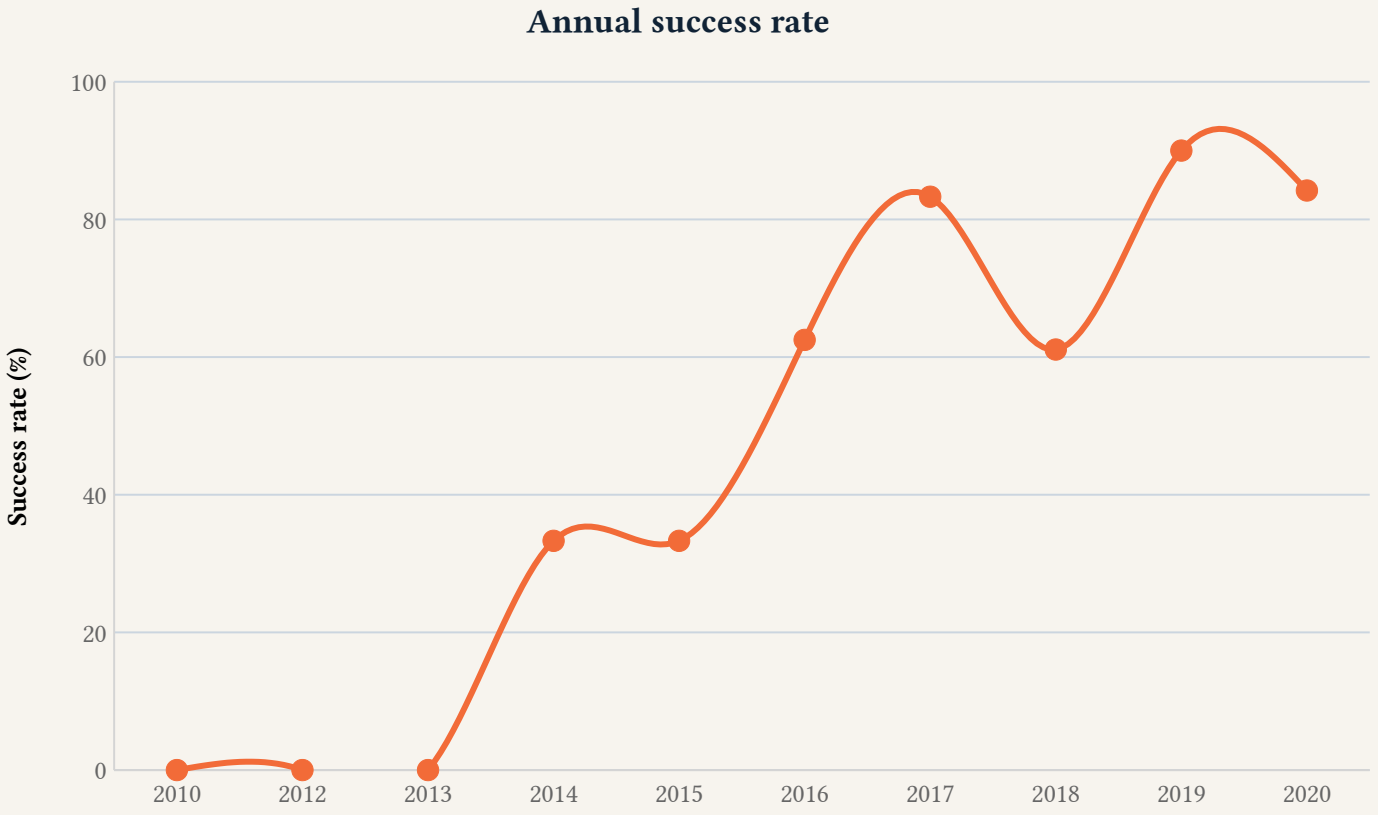
Site is informative, but sample sizes differ.

Payload mass alone does not determine landing outcome



- Successes span light and heavy payloads.
- High-mass VLEO launches are mostly successful.
- Mission era, orbit, and site add context.

Landing performance improved over time



Orbit profile

Orbit	Launches	Success
SSO	5	100.0%
VLEO	14	85.7%
LEO	7	71.4%
PO	9	66.7%
ISS	21	61.9%
GTO	27	51.9%

2019 peaked at 90.0%; 2020 remained high at 84.2%.

SQL results — sites, payloads, and outcomes

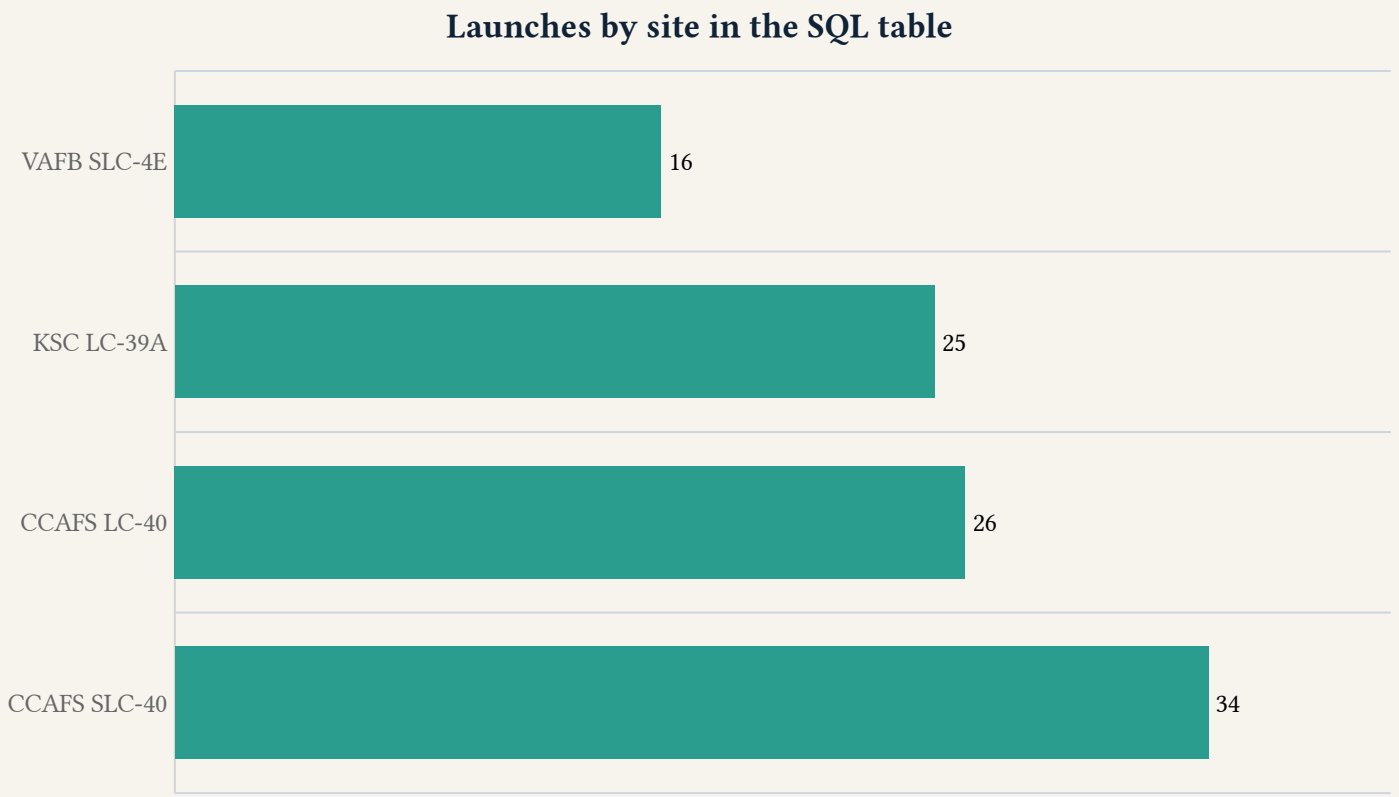
Question	Result
Distinct launch sites	4
NASA CRS payload mass	48,213 kg
First successful ground-pad landing	2015-12-22
Mission outcomes marked Success	98 of 101
2015 drone-ship failures	2

4

successful drone-ship landings with payloads between 4,000 and 6,000 kg

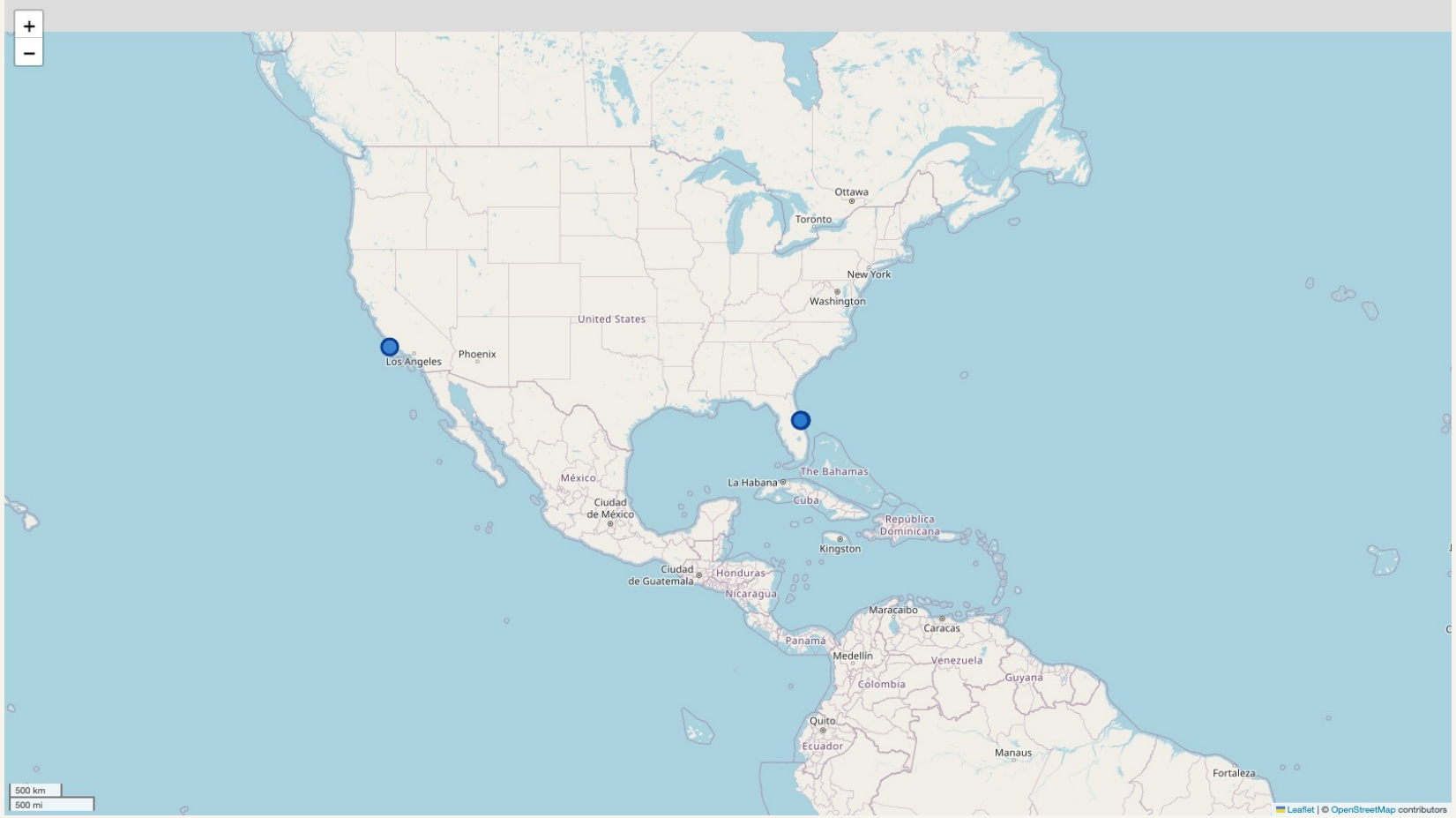
JCSAT-14 · JCSAT-16 · SES-10 · SES-11/EchoStar 105

SQL results — launch-site ranking



- CCAFS SLC-40 ranks first with 34 launches.
- Twelve boosters share the maximum payload mass of 15,600 kg.
- SQL answers are generated from the included 101-row table.

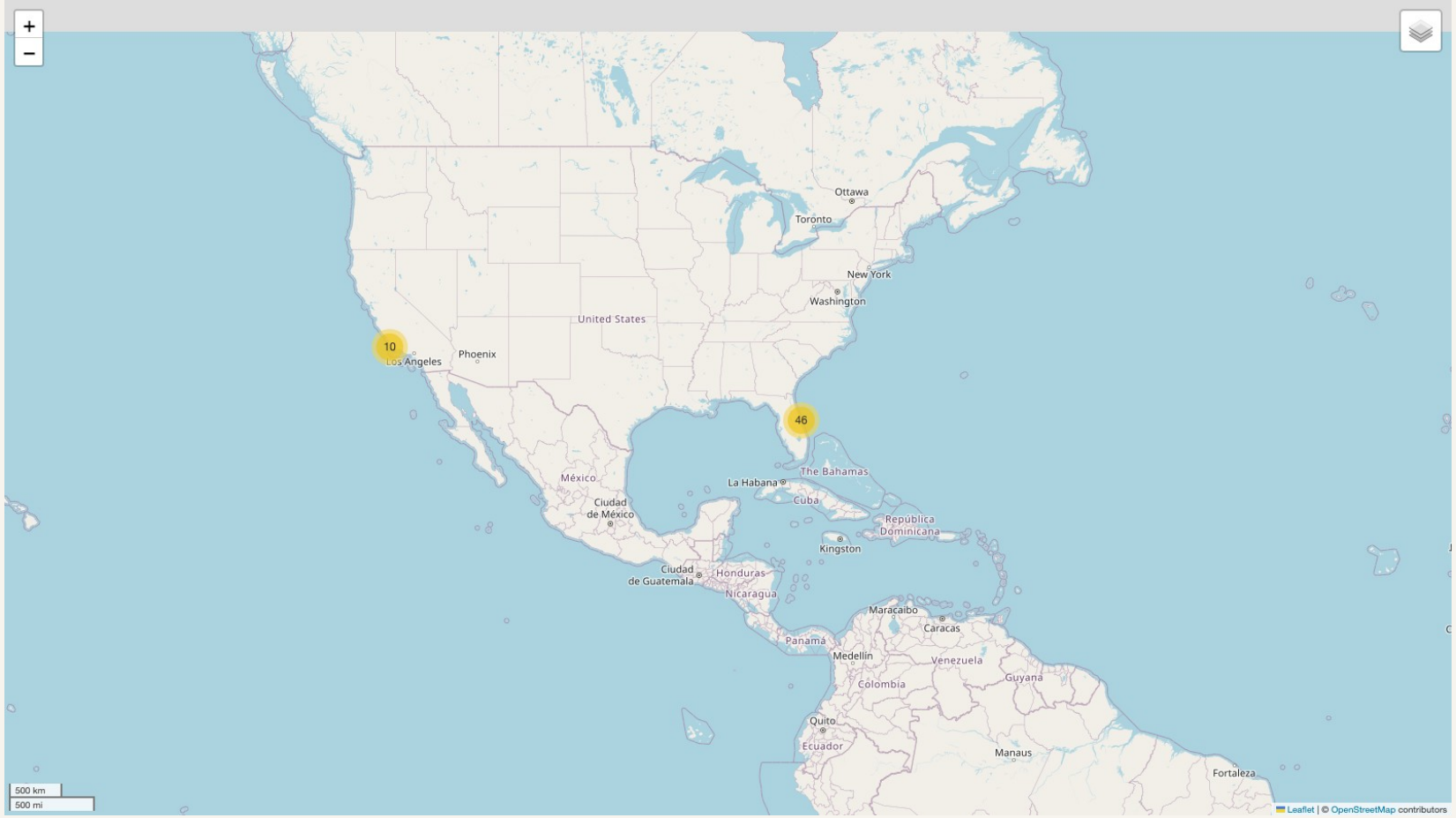
Folium — launch-site markers



- Three Florida pads cluster around Cape Canaveral / Kennedy Space Center.
- VAFB SLC-4E is on the California coast.
- Marker popups retain site name and coordinates.

INTERACTIVE ANALYTICS

Folium — launch records and outcomes



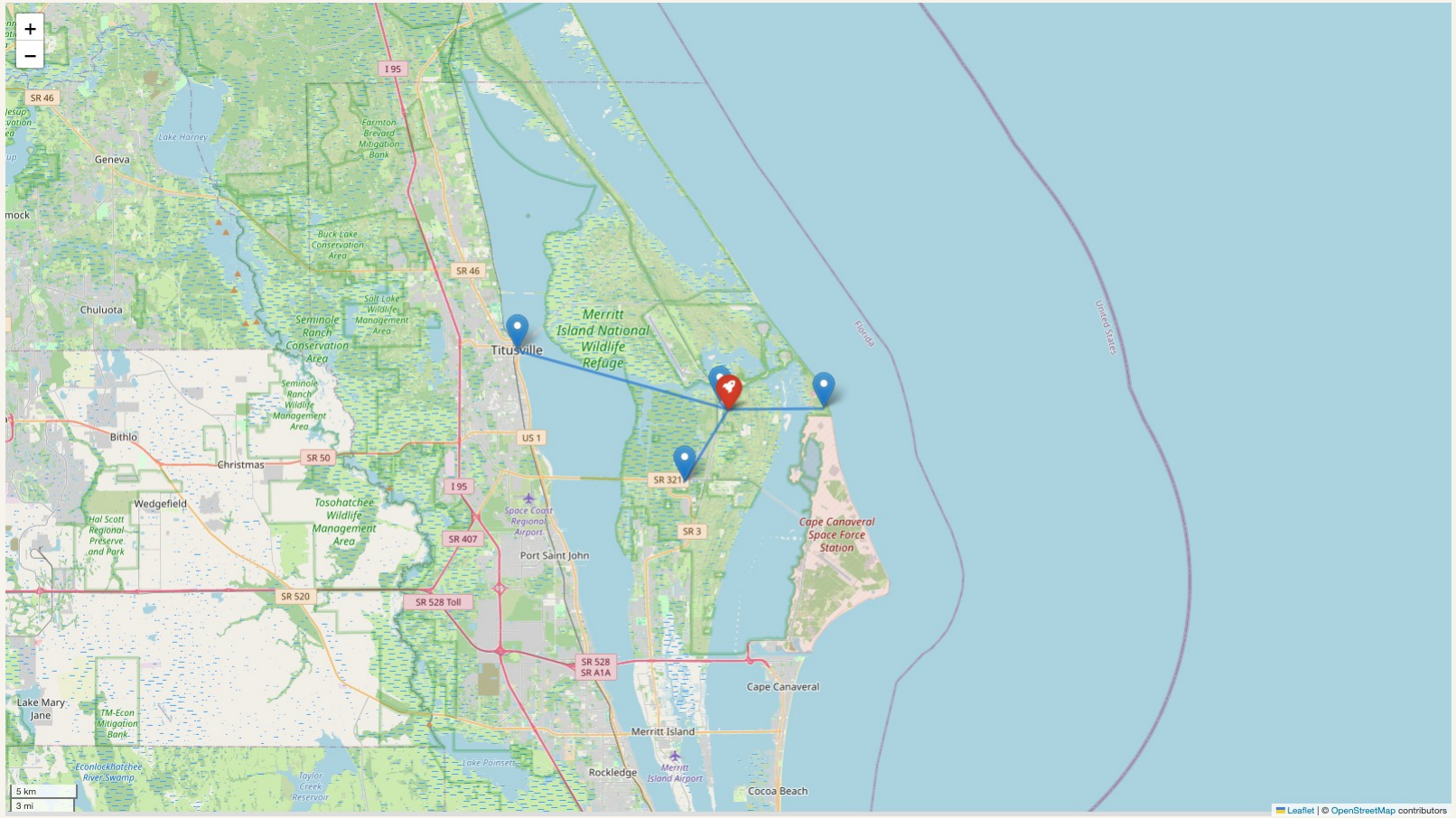
46

mapped launch records clustered in Florida

10

mapped launch records clustered in California

Folium — proximity to infrastructure



7.0 km

Atlantic coastline

0.9 km

NASA railway

6.3 km

NASA Causeway

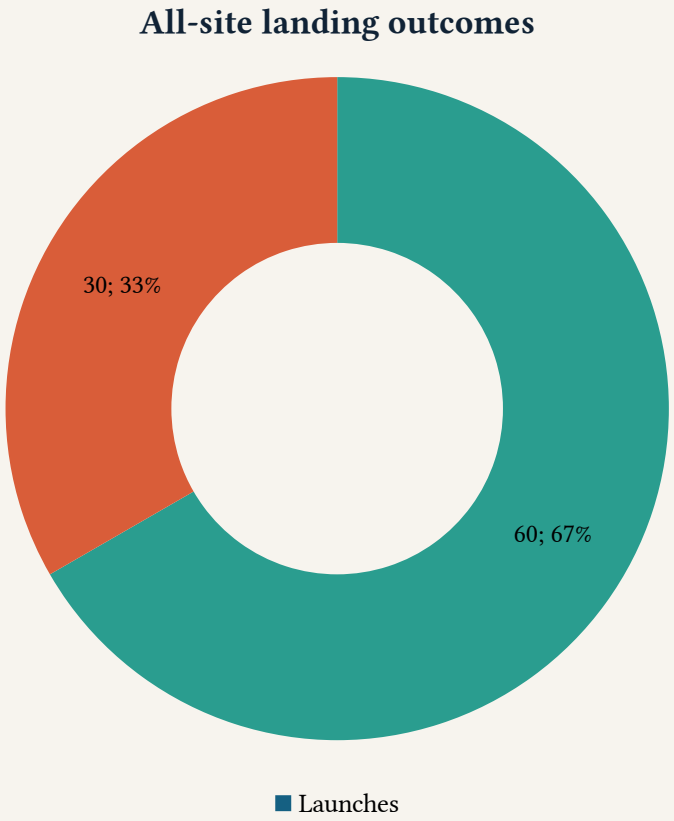
16.3 km

Titusville

Nearby rail and road access support logistics; coastal siting also separates launch trajectories from populated areas.

INTERACTIVE ANALYTICS

Dash — success share by site

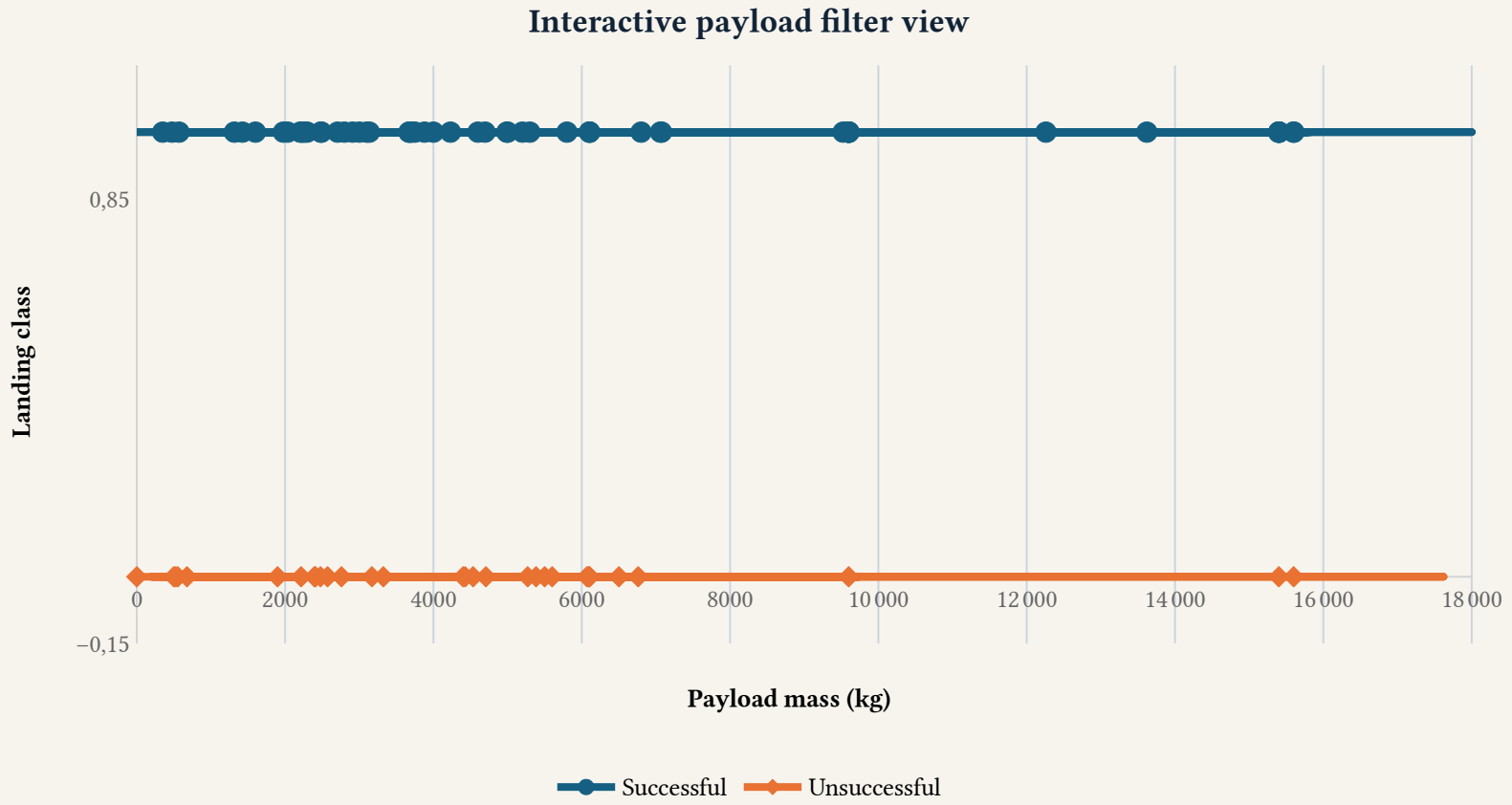


Dashboard controls

- Site dropdown: all or one launch site
- Payload range: minimum to maximum mass
- Pie and scatter update from the same filter

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Dash — payload versus outcome



Why interactive?

Users can isolate a site and payload band, then inspect how booster categories and outcomes change together.

PREDICTIVE ANALYSIS

Predictive modeling methodology

Logistic regression

linear baseline

SVM

nonlinear decision boundary

Decision tree

rule-based splits

KNN

local neighborhood

1

Split

72 train / 18 test

2

Scale

inside each pipeline

3

Tune

GridSearchCV

4

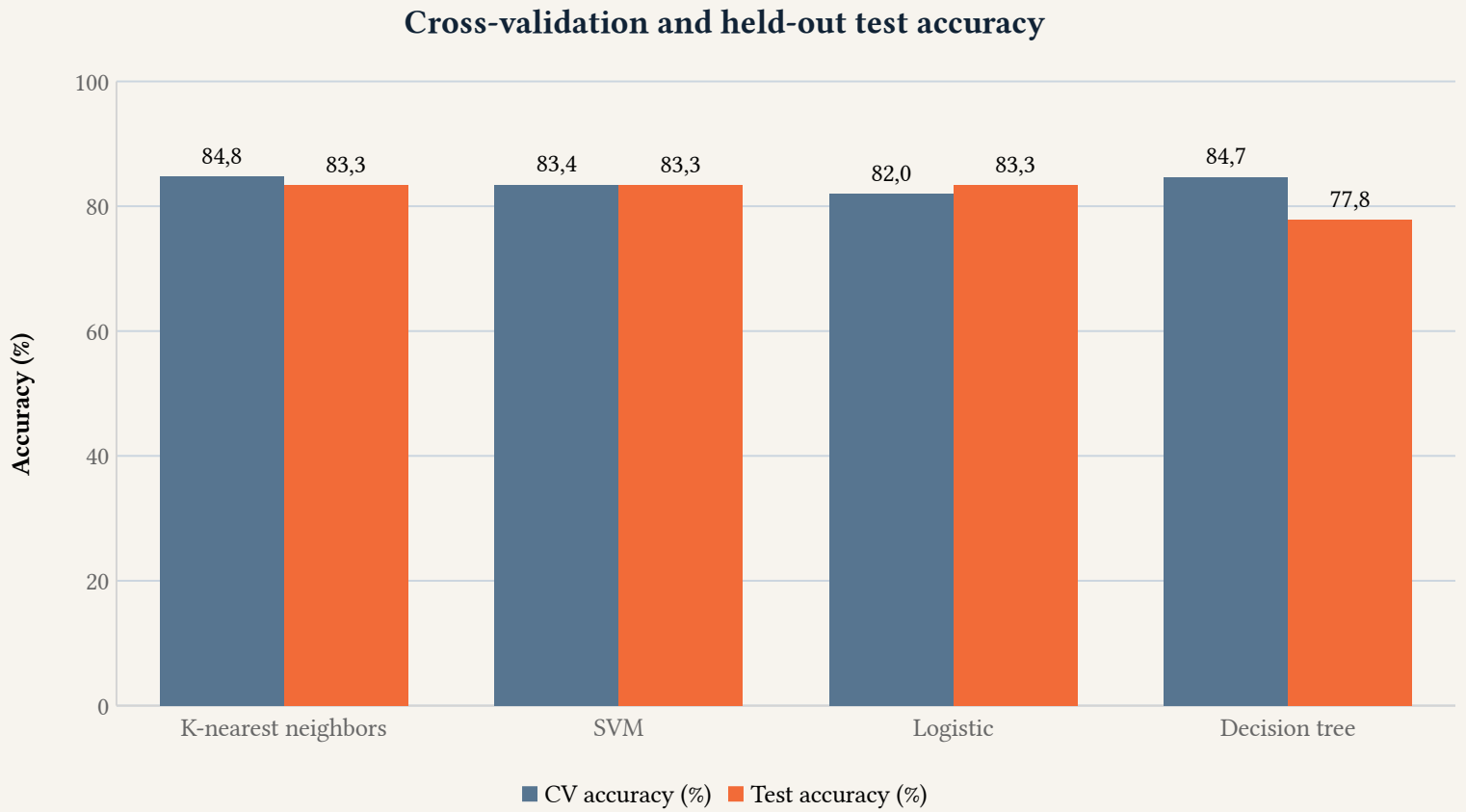
Evaluate

accuracy + precision/recall/F1

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PREDICTIVE ANALYSIS

Three models tie at 83.3% test accuracy



Selected model

K-nearest neighbors

Highest CV accuracy
84.8%

Best-model confusion matrix

Actual / Predicted	0	1
0 — unsuccessful	3	3
1 — successful	0	12

12
true positives

3
true negatives

3
false positives

0
false negatives

Recall is 100%, but the model over-predicts success for half of the actual failures in this small test split.

CONCLUSION

Conclusions and next steps

What the evidence supports

- Landing success rose materially after 2015.
- Site, orbit, and mission era carry predictive signal.
- KNN is the CV winner, not a uniquely dominant test winner.

What comes next

- Use time-aware validation to prevent era leakage.
- Refresh the dataset with newer missions and richer weather/trajectory features.
- Calibrate probabilities before any cost or risk decision.

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